Yuheng Zhu

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Education

North Carolina State University — *Aug 2024 – Present*
Ph.D. in Computer Science

North Carolina State University — *Aug 2022 – May 2024*
Master in Computer Science

Southern University of Science and Technology — *Sept 2017 – May 2021*
B.E. in Computer Science and Technology

Experience

Engineer Intern — *San Diego, CA*
Qualcomm — *Jun 2025 - Aug 2025*

Research Assistant — *Raleigh, NC*
North Carolina State University — *Sept 2023 - Present*

Research Assistant — *Shenzhen, China*
Southern University of Science and Technology — *June 2021 – July 2022*

Software Engineer Intern — *Chengdu, China*
JD.com, AI Research Center — *June 2020 – Aug 2020*

Publications

AdaptAV: Continuous Adaption of Vision Models for Autonomous Vehicles Using Cloud-based Oracle. ***Yuheng Z.***, Dhruva U., Boluo G. & Man-Ki Y., *Proceedings of the 100th IEEE Vehicular Technology Conference*

Bridging Data and Knowledge: A Neurosymbolic Framework for Reliable Network Analysis. Zhjin Y., ***Yuheng Z.***, Mingzhe C. & Yuchen L., *Proceedings of 2025 IEEE Global Communications Conference*

FrameScope: Temporal Data Valuation for Stream Active Learning in Autonomous Vehicle Systems. ***Yuheng Z.***, & Man-Ki Y. *Proceedings of The 10th ACM/IEEE Symposium on Edge Computing*

RinneFormer: Transformer-based Real-world Cooperative Perception Algorithm. Deveshwar H., ***Yuheng Z.***, Dhruva U., Man-Ki Y. & Seth H. *Proceedings of 2026 IEEE Intelligent Vehicles Symposium*

Projects

Transformer-based Cooperative 3D Object Detection — *NCSU, 2025*

- Designed a Transformer-based V2X detection framework featuring Point Transformer and Cross Attention modules to capture fine-grained geometry within pillars to resolve sparse LiDAR geometry, achieving SOTA accuracy and superior generalization in real-world cooperative datasets.
- Engineered a distributed ROS multi-agent pipeline robust to asynchronous data; validated system deployability and latency stability using raw sensor streams from physical autonomous vehicles.

Agentic LLM Testcase Triage Framework — *Qualcomm, 2025*

- Independently built an agentic RAG testing failure analysis pipeline for 10K+ tests/day.
- Multi-source logs distilled via LLM summarization + error-anchor context, then chunked and indexed in a vector storage (ChromaDB).
- Orchestrated with LangChain agents + function calling for hybrid retrieval (semantic + keyword) and top-k reranking, emitting JSON-schema reports.
- Tools used: **LangChain, ChromaDB, FAISS**

On-the-fly Coding of Vision Inputs for Evidence-Preserving Perception — *NCSU, 2023 - 2024*

- Designed and implemented a visual perception middleware for autonomous driving systems that preserves complete and reproducible evidence of visual inputs.
- Utilized compression techniques such as JPEG and H.264, with performance optimizations achieved through NVENC and NVDEC hardware acceleration engines.
- Tools used: **C, Python, V4L2, , NVIDIA Jetpack, NVIDIA Codec, FFmpeg**

Carla1s Distributed Online Simulation Platform — *SUSTech, 2021 - 2022*

- Designed and developed a Carla-based remote and hardware-free web simulation platform.
- Developed a **backend streaming server** for **HTTP Streaming ROS** Image Topics and Carla Sensor Output, supporting multiple video encodings and streaming protocols, reduce latency of remote simulation

from 1-2s to 50ms.

- Tools used: **Python Flask, FFmpeg, Ansible, Git, Docker**

Jingxiaozhi AI Customer Service System *JD.com, 2020*

- Participated in two development cycles (2.1.6 & 2.1.7) of JD Jingxiaozhi online AI customer service system, made in-depth research on the application and practice of Chinese **NLP** in e-commerce scenarios
- Developed web API automated testing modules in **Java** based on JD JSF **RPC framework**, made test framework can fully mimic user operation logic, increased API test coverage to 100%
- Tools used: **Java, RESTful API, JavaScript, Redis, RabbitMQ**

Technologies

Languages: C, C++, Shell, Python, Java, JavaScript, Lua

Skills: Linux Kernel, ROS2, Networking, DevOps, Docker, Redis, Carla, ADAS, LangChain, ChromaDB